Interval level of measurement has fixed intervals or scales so you can add and subtract.

Interval scales don’t have a “true zero” but have fixed intervals.

Calendar years, Celsius, and Fahrenheit temperature scales are common examples.

We can say 10° degrees C is 8° degrees hotter than 2° degrees, but it isn’t 5 times as hot. Or 40° degrees F is 20° hotter than 20° F but it isn’t twice as hot.

The year 1000 is a thousand years before 2000 but it isn’t half the time as the year 2000 in the general length of time.

2010 is ten years before 2020 but we can’t talk about a ratio here.

Ratio scales or ratio data, has an “absolute zero” and you can do multiplication and division.

Elapsed time is a ratio scale since it starts at a certain time that represents zero.

Height, weight, length, age, and others are all ratio scale when they start at zero.

Kelvin is the temperature scale that starts at absolute zero. So, 8 K is twice as hot as 4 K.

Age is a scale that starts with zero as birth, as opposed to calendar year.

If Doc is 4 and Aury is 2 then Doc is twice as old as Aury.

Historically new types of number systems get added.

Counting numbers: 1,2,3…

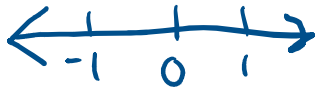
Natural numbers: counting numbers with zero $0, 1, 2, 3 \ldots$

Integers: positive and negative counting numbers $\ldots -2, -1, 0, 1, 2 \ldots$

Rational numbers: can be written as a fraction of two integers $\ldots -\frac{1}{2}, 0, \frac{1}{2}, \frac{2}{2}, \frac{3}{2} \ldots$

Irrational numbers: cannot be expressed as a fraction π $\sqrt{2}$ e

Real numbers: include all the rational and irrational numbers



Complex numbers: include the imaginary numbers $3 + 2i$

$\sqrt{-1}$ is i

Surreal numbers: include infinitesimals

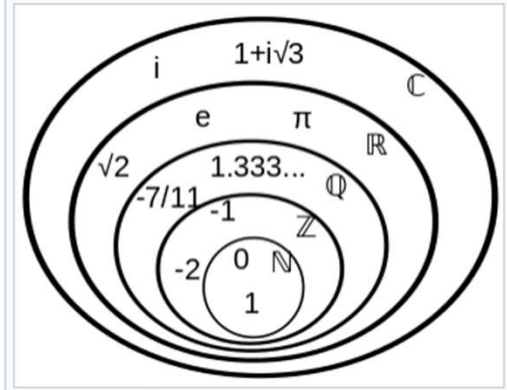
δ is an infinitely small number

A is a **subset** of B

There are sets of such mathematical importance, to which mathematicians refer so frequently, that they have acquired special names and notational conventions to identify them.

Many of these important sets are represented in mathematical texts using bold (e.g. $\mathbf{Z}$) or blackboard bold (e.g. $\mathbb{Z}$) typeface.[38] These include:[14]

- $\mathbf{N}$ or $\mathbb{N}$, denoting the set of all natural numbers: $\mathbf{N} = \{0, 1, 2, 3, ...\}$ (some authors exclude 0)[38]
- $\mathbf{Z}$ or $\mathbb{Z}$, denoting the set of all integers (whether positive, negative or zero): $\mathbf{Z} = \{..., -2, -1, 0, 1, 2, ...\}$[38]
- $\mathbf{Q}$ or $\mathbb{Q}$, denoting the set of all rational numbers (that is, the set of all proper and improper fractions): $\mathbf{Q} = \{a/b \mid a, b \in \mathbf{Z}, b \neq 0\}$, for example, $1/4 \in \mathbf{Q}$ and $11/6 \in \mathbf{Q}$, and since every integer n can be expressed as the fraction $n/1$, all integers are members of this set ($\mathbf{Z} \subsetneq \mathbf{Q}$)[38]
- $\mathbf{R}$ or $\mathbb{R}$, denoting the set of all real numbers, including all rational numbers, together with all irrational numbers (that is, algebraic numbers that cannot be rewritten as fractions such as $\sqrt{2}$, as well as transcendental numbers such as π, e)[38]
- $\mathbf{C}$ or $\mathbb{C}$, denoting the set of all complex numbers: $\mathbf{C} = \{a + bi \mid a, b \in \mathbf{R}\}$, for example, $1 + 2i \in \mathbf{C}$[38]



The natural numbers ℕ are contained in the integers ℤ, which are contained in the rational numbers ℚ, which are contained in the real numbers ℝ, which are contained in the complex numbers ℂ

Each of the above sets of numbers has an infinite number of elements, and each can be considered to be a proper subset of the sets listed below it.

Sets of positive or negative numbers are sometimes denoted by superscript plus and minus signs, respectively. For example, $\mathbb{Q}^+$ represents the set of positive rational numbers.

Set (mathematics) - Wikipedia

With algebra, we usually use the set of Real numbers.

Counting numbers: 1,2,3…

Natural numbers: counting numbers often with zero

Integers: positive and negative counting numbers $\ldots -2, -1, 0, 1, 2 \ldots$

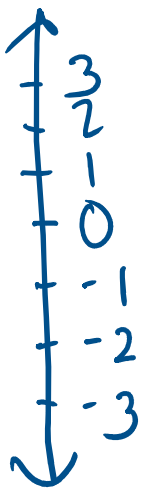
Rational numbers: can be written as a fraction of two integers $\frac{1}{2}$ or $\frac{2}{1}$

Irrational numbers: decimals that go on forever $\pi \approx 3.14\ldots$

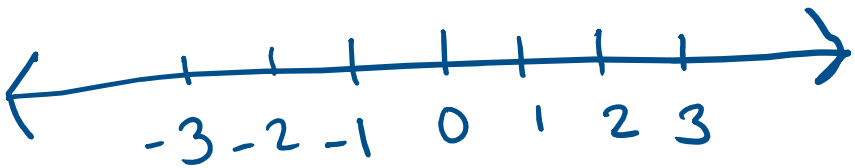
Real numbers

$\mathbb{R}$

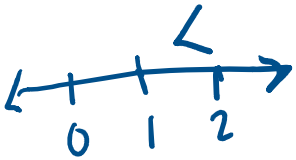
We can represent the Real numbers on a number line. On a vertical number line, the positive numbers get bigger going up and the numbers get more negative going down. This is easier for people like me who confuse Right and Left.



A horizontal number line has zero in the middle. The positive numbers get bigger to the right, and the negative numbers get bigger to the left.



Warning! The base 10 number system writes the larger numbers to the left, but the number line has the larger numbers to the right. With my brain wiring I get my right and left confused. If this is an issue for you, there is a trick. Left, L less than.

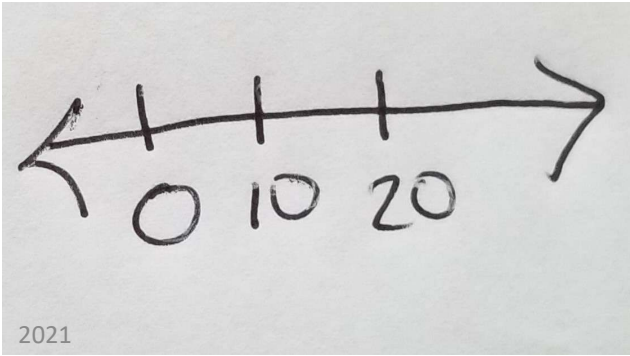


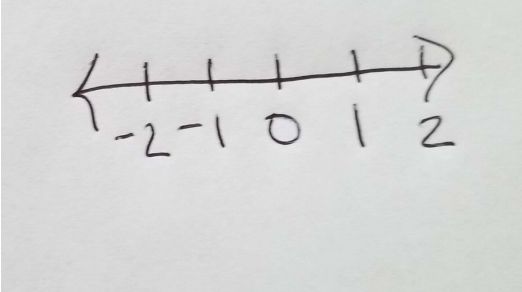
Remember: horizontal number lines get greater to the right.

In our decimal system, bigger numbers are to the left, and we write from greatest or biggest to least or smallest.

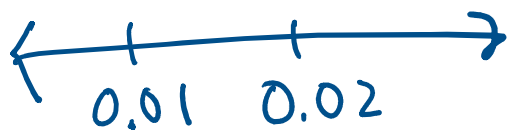
2139.04 the 2 represents two thousand which is much bigger than the 4 which represent 4 hundredths or $\frac{4}{100}$.

Caution! We are trained to think of the bigger numbers to the left and then the number line has the greater numbers to the right!



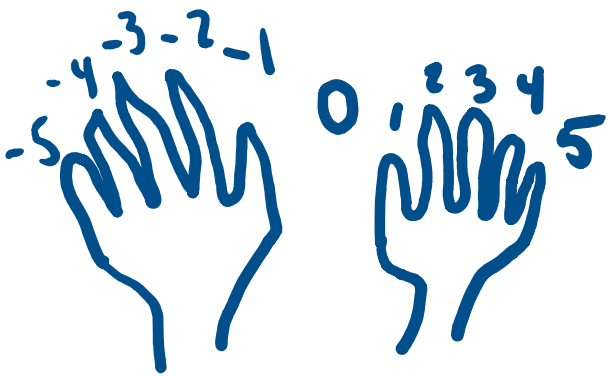


A number line can be zoomed in or zoomed out and you can show a piece of a number line. I will usually show the integers right around zero as that is much easier for the brain to process.



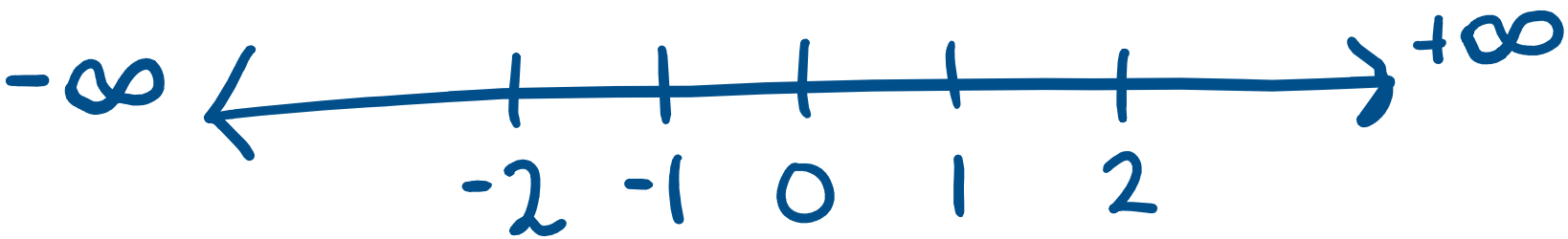
You can use your fingers to practice integers in the direction of the number line.

(I think I need to add a photo of human hands as these look alien.)



On a number line there is infinity in both directions, and we can refer to positive infinity and negative infinity.

$+\infty$ $-\infty$

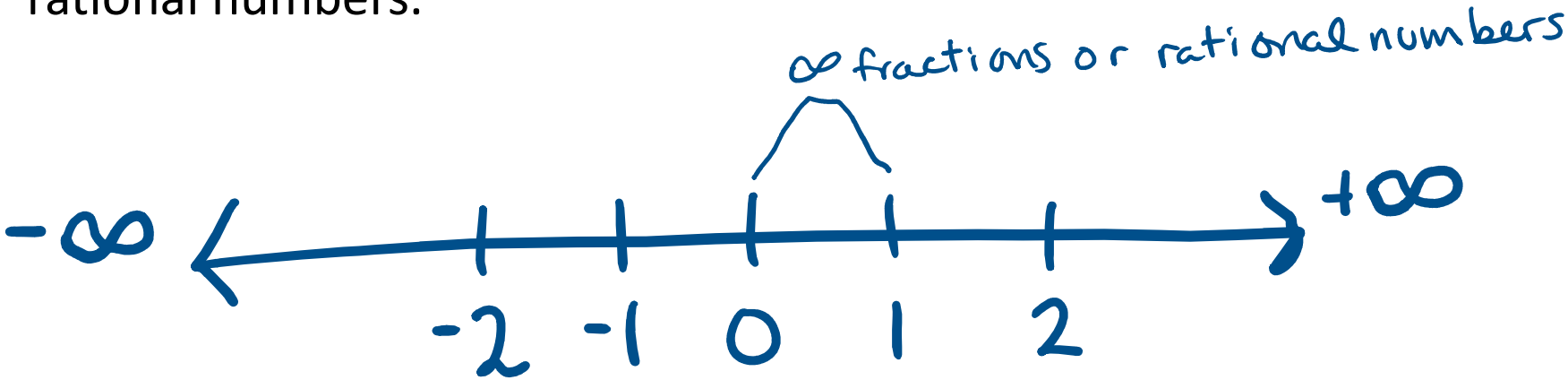


The concept of infinity can be hard for some people to grasp.

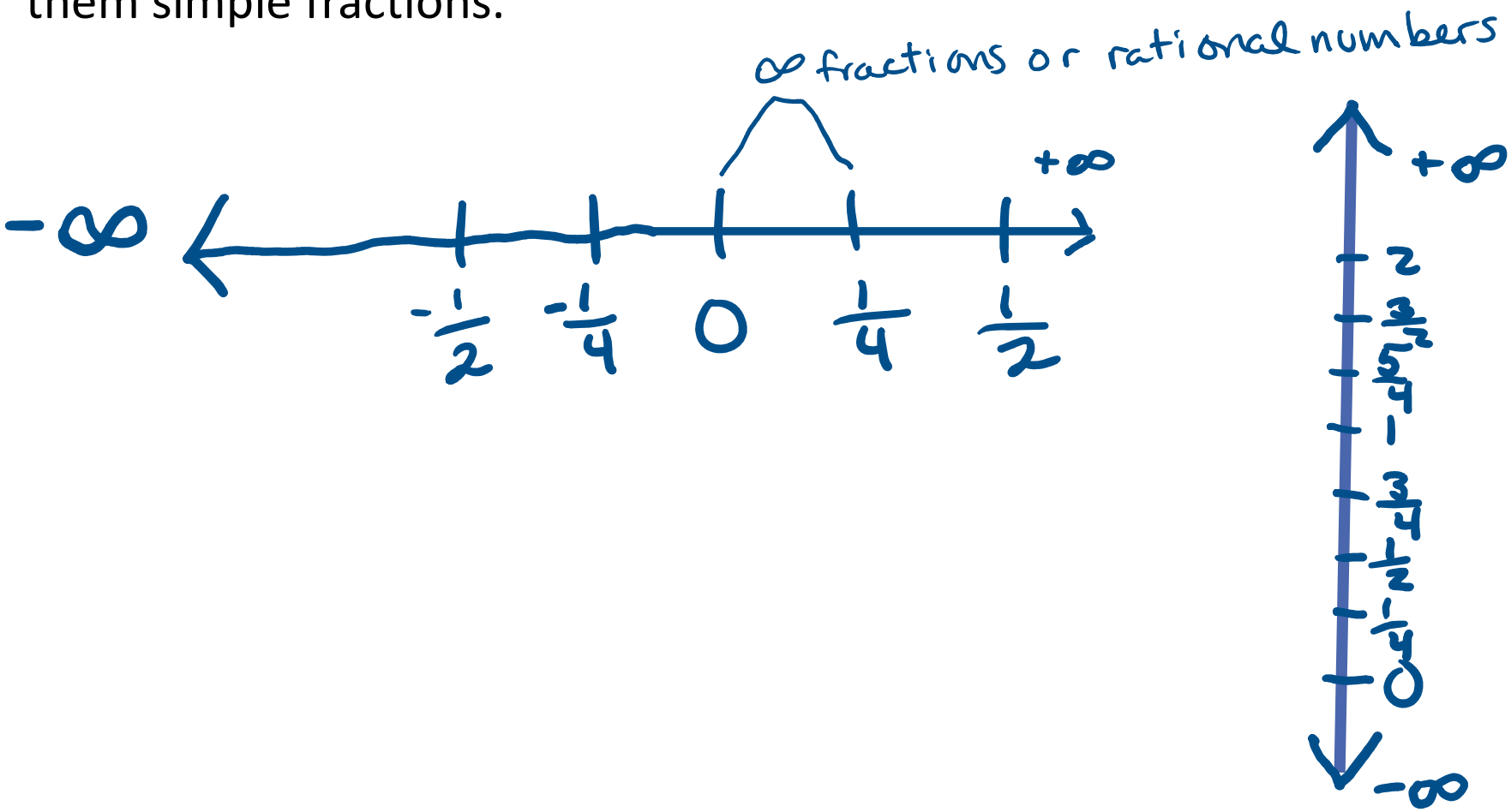
I like to use this video to show the concept of getting infinitely large and small. Of course, the video is finite (meaning it has a beginning and end.) ☺

Universe Size Comparison | Cosmic Eye (Original HD)

In between any two of the numbers shown, there are infinitely many rational numbers.



Rational numbers can be written as a ratio of two integers. We also call them simple fractions.



In between any rational numbers, there are infinitely many irrational numbers. These can't be written as a fraction or a ratio of integers.

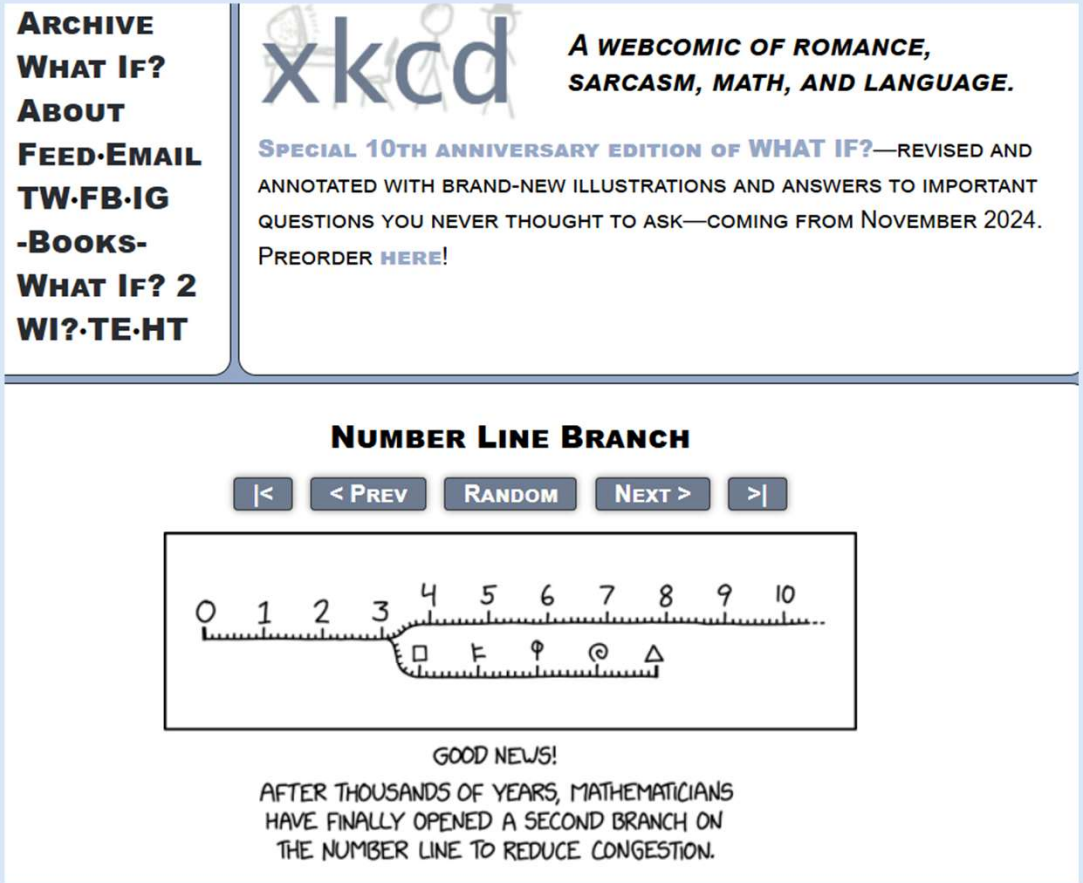
We will talk about these irrational numbers as they come and the ones that we care about are pi, the square roots, other roots, and e.

π $\sqrt{2}$ $\sqrt{5}$ $\sqrt[3]{10}$ $e \approx 2.7$

Because we can't write them as a fraction and they keep going on infinitely as decimals, we have special names and symbols for them.

You may have seen pi and square roots.

e (mathematical constant) – Wikipedia is the base for the exponential function which is the fixed point for the derivative and integral in calculus, so we will learn about *e* then.

Now we will move into operations on numbers and variables. But first, a little math comic.

https://xkcd.com/2956/